

ATLS — Advanced Trauma Life Support (Revision Table)

Bold = high-yield. 3 steps: Primary survey (+ resuscitation) → Secondary survey (head-to-toe) → Definitive care. Team led by most senior clinician.

1 — PRIMARY SURVEY = cABCDE (identify + treat what is killing the patient)

Step	Assessment	Key action / pearls
c — Catastrophic haemorrhage	Control massive EXTERNAL bleeding FIRST	Direct pressure, tourniquet
A — Airway + C-spine	Verbal response = airway OK. No response → assess obstruction (foreign body, maxillofacial/mandibular #, laryngeal injury, oedema)	Chin lift / jaw thrust. Protect C-spine throughout. Normal neuro does NOT exclude C-spine injury
B — Breathing	High-flow O2 (reservoir mask). Assess surgical emphysema, neck veins, symmetry, effort. Percuss + auscultate front/back	6 clinical (NOT radiological) life-threats: tension pneumo, flail chest+contusion, massive haemothorax, open pneumo (+ airway, tamponade)
B — chest treatment	Pneumo/haemo → tube thoracostomy (chest drain). Tension pneumo = EMERGENCY → needle decompression (mid-clavicular line, 2 finger-breadths below clavicle) then chest drain	Hyperresonance=tension; dullness=haemothorax; tracheal deviation
C — Circulation	3 obs: conscious level (↓=blood loss), skin (pale/ashen/grey=hypovolaemic), pulse (rapid thready/impalpable=shock). Pelvic binder, don't remove till # excluded	TXA ASAP. Resuscitate with BLOOD/products, NOT crystalloid. Crystalloid = small amounts to maintain BP (permissive hypotension). Massive transfusion protocol
D — Disability	GCS + pupils + motor. GCS = E+V+M, best 15, worst 3	Exclude hypoglycaemia, alcohol, drugs
E — Exposure	Full expose, examine front+back via controlled log roll with in-line traction	Prevent hypothermia → warming air blankets

2 — SECONDARY SURVEY (head-to-toe, after primary + resuscitation started)

Imaging	Whole-body CT (WBCT) head→pelvis + IV contrast = GOLD STANDARD for severe blunt trauma. No selective scanning; time-critical/early. If immediate laparotomy without WBCT → pelvic binder + pelvic X-ray
Head / face	Maxillofacial #, ocular injury, open head injury. Ear bleeding/discharge = basal skull fracture. Inspect mouth, mandible, zygoma, nose, ears
Neck	Palpate C-spine (haematoma, crepitus, steps). Assume injury if uncertain/significant head injury → hard collar + sandbag + tape. Explore wounds deeper than platysma in theatre
Chest	Palpate clavicle/sternum/ribs. Sternal # → cardiac damage risk → monitor 24–48h. Distended neck veins + distant heart sounds + narrow pulse pressure = cardiac tamponade (Beck's triad)
Neuro / Extremities	GCS every 15 min. Sensory/motor deficit → spinal immobilization + neurosurgery. Limbs: align deformities (document neurovascular before + after), move joints (dislocations)
Log roll	≥ 4 people : 1 in-line spinal traction (leads/calls — anaesthetist if intubated), 1 torso, 1 pelvis/legs, 1 removes board + examines back (occiput→sacrum, penetrating/exit wounds, percuss/auscultate)

3 — DEFINITIVE CARE: ETC vs DCS & LACTATE

ETC (Early Total Care)	Definitive management of ALL injuries within 36h after initial resuscitation. For physiologically stable patient
DCS (Damage Control Surgery)	Simultaneous resuscitation + early rapid life/limb-saving surgery ; defer time-consuming definitive surgery until physiology allows. ETC → switch to DCS if deteriorates
Venous lactate (resuscitation marker)	<2 → ETC · 2–3 → watch trend · >3 → under-resuscitated (more resus or DCS if urgent) · >5 → DCS